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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – JULY / AUGUST 2025

Program	:	**B.E. – CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)**	**Semester**	:	**IV**
Course Name	:	**Design and Analysis of Algorithms**	**Max. Marks**	:	**100**
Course Code	:	**22 / 23CY43 / 22 / 23CY43**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) Discuss the asymptotic bounds for algorithms with an example. CO1 (07)

   b) Apply Linear Search algorithm to search the list 10,92,38,74,56,19,82,37 for a key value 74. Calculate the time complexity in Best Case and Worst case inputs for the algorithm. CO1 (07)

   c) Examine the time complexity of the following: CO1 (06)

   i)
```
int fun(int n)
{
  int count = 0;
  for (int i = 0; i < n; i++)
     for (int j = i; j > 0; j--)
        count = count + 1;
  return count;
}
```

   ii)
```
void fun(int n, int arr[])
{
    int i = 0, j = 0;
    for(; i < n; ++i)
        while(j < n && arr[i] < arr[j])
           j++;
}
```

2. a) Illustrate the following with respect to Gale-Shapley algorithm CO1 (08)
   i. Perfect Matching
   ii. Stable Matching
   iii. Man Optimal
   iv. Woman Optimal.

   b) Consider the scenario given. CO1 (06)
   There are n people numbered from 1 to n in a circle. Starting from person 1, we eliminate every second person until only survivor is left.
   - Design an efficient algorithm to find the survivor's number J(n).
   - Calculate the time complexity class for the algorithm you defined.

   c) Solve the following recurrence relations using substitution method CO1 (06)
   i. x(n)=x(n-1)+5 for n>1,x(1)=0
   ii. x(n)=3x(n-1) for n>1, x(1)=4

UNIT - II

3. a) Design and Implement a Merge Sort algorithm to sort a given set of elements list = { 5, 8, 2, 7, 4, 1, 6, 3 } and determine the time required to sort the elements. Prove that the time complexity of merge sort is O (n log n) using "Masters theorem" and deduce the recurrence relation for the same. CO2 (08)

b) Examine the BFS traversal for a graph given in Fig. 3(b) considering the start vertex as 'A'. CO2 (06)

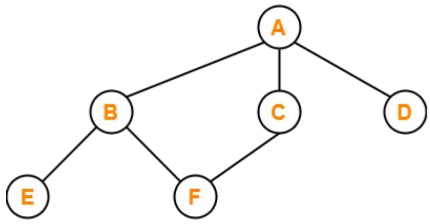


Fig. 3(b) Graph

c) Describe an algorithm for Topological sorting using Directed Acyclic Graph approach. CO2 (06)

4. a) Design and Implement Quick sort algorithm to sort a given list of unsorted elements in ascending order and determine the time required to sort the elements. Calculate the Time complexity for the algorithm. List = { 10, 20, 30, 40, 50, 60, 70 }. CO2 (06)

b) Construct the DFS Traversal tree for the graph given in Fig. 4(b) Graph with "A" as the source vertex using Stack and Write the algorithm for the same. CO2 (08)

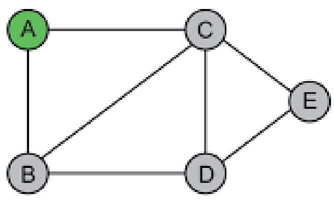


Fig. 4(b) Graph

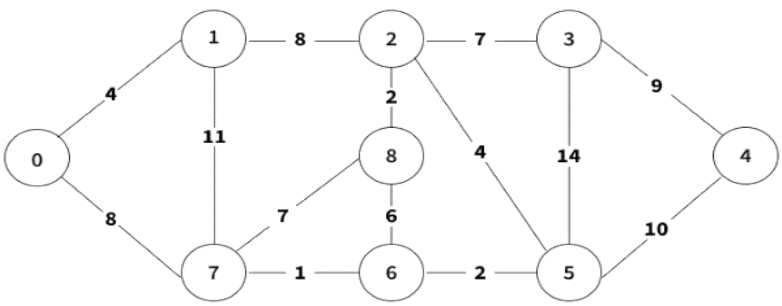
c) Write the Recurrence Relation for Merge-Sort Algorithm. Analyze the running time of Merge-Sort for both best and worst case. CO2 (06)

UNIT - III

5. a) Write the heap sort algorithm. Sort 32, 84, 62, 33, 94, 80, 24 using heapsort. CO3 (10)

b) Illustrate the greedy technique with appropriate examples. Write the pseudocode of Kruskal algorithm. CO3 (10)

6. a) Apply prim's and Kruskal's algorithm to find the minimum spanning tree (MST) of the following graph: CO3 (10)



b) Write the stepwise algorithm for constructing Huffman trees. Construct Huffman Tree for the following data. Also encode the text "DAD" and decode the codeword "10011011011101". CO3 (10)

symbol	A	B	C	D	_
frequency	0.35	0.1	0.2	0.2	0.15

UNIT- IV

7. a) Find the shortest path from node 1 to every other node in the graph given in Fig. 7(a) using Bellman-Ford algorithm. CO4 (08)

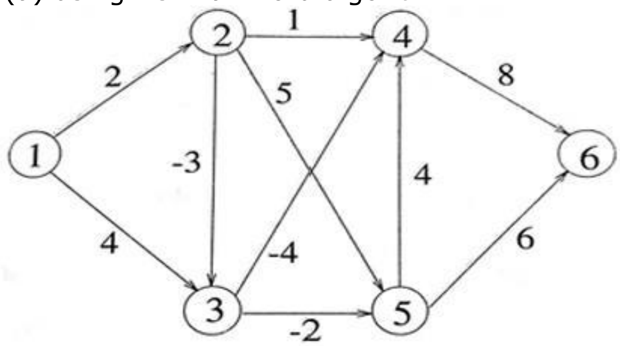


Fig. 7(a) Graph

b) Explain the steps for solving Dynamic programming problems. CO4 (06)

c) Discuss the weighted interval scheduling problem with an example. CO4 (06)

8. a) Describe the working of Maximum flow problem with an example. CO4 (08)

b) Solve the Knapsack instance with the given input data by identifying the items to be placed in the knapsack with capacity W=10 and write an algorithm for the same: CO4 (06)

Item	Weight	Value
1	7	42
2	3	12
3	4	40
4	5	25

c) Explain the general principles of Dynamic Programming. CO4 (06)

UNIT - V

9. a) Explain the following with appropriate examples: CO5 (10)
   i) Polynomial Time Reductions
   ii) NP-Complete Circuit Satisfiability

b) Summarize the following concepts: CO5 (10)
   i) A First NP-Complete Problem
   ii) General Strategy for Proving New Problems NP-Complete

10. a) Outline the following with suitable examples: CO5 (10)
   i) Sequencing Problems
   ii) The Travelling Salesman Problem

b) Explain the Hamiltonian Cycle Problem with appropriate examples. CO5 (10)

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